# OMAR CUSMA FAIT

**Data Science and Artificial Intelligence Student at the University of Trieste**


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## ABOUT ME

I possess a strong background in **mathematics** and **physics**, which has refined my **analytical thinking** and **problem-solving** skills. These abilities have proven invaluable in approaching complex challenges from a logical perspective.

## WORK EXPERIENCE

### Math Teacher

**ISIS Gregorčič**

March 2025 – June 2025 | Gorizia

- Taught mathematics at the high school level, preparing and delivering lessons, supporting students in building solid problem-solving foundations, and assessing progress across the school term.

## INTERESTS

Python | Mathematics | Physics | Science | Simulations | Reinforcement Learning | AI | Machine Learning | Big Data | Deep Learning

## REFEREES

**Dr. Elena Orlando**
INAF
orland.ele@gmail.com

**Marco Zennaro**
Internship supervisor
Available upon request

**Alessio Ansuini**
Thesis supervisor
Available upon request

**Peter Černic**
Principal, ISIS Gregorčič – Gorizia
Available upon request

## DIGITAL SKILLS

Python | Programming | Agentic AI | Pandas | NumPy | MatPlotLib | GitHub | PyCharm | Latex | Krita | Windows | Excel | Word | Pascal | PowerPoint

## LANGUAGES

**Italian (native)** ●●●●●

**Slovenian (native)** ●●●●◐

**English (B2)** ●●●●◐

Learning: Japanese

## EDUCATION

### Master's Degree in Data Science and Artificial Intelligence

**Università degli Studi di Trieste – DSAI**

Thesis title: Unsupervised State Bucketing for Mixture of Experts in Resource-Constrained Chess Engines

### Bachelor's Degree in Physics

**Università degli Studi di Trieste**

Thesis title: A Comparative Analysis of Neural Network Methods for Source Detection

### High School Diploma Applied Sciences

**Istituto di Istruzione Superiore Gabriele D'Annunzio**

# PROJECTS

## Astrophysics Internship

2023 INAF

- In a Study of the local correlation between two HEALPix images, I faced an open-ended problem for which I had to come up with my own solution to the problem.

## AI internship

2026 ICTP

- I developed a data-driven Mixture-of-Experts architecture to optimize resource-constrained NNUE chess engines through gradient-informed sample clustering and a lightweight inference dispatcher.

## Semantic Retrieval System with LLMs

- A system that can search through a folder of documents to find the most relevant one, highlight the most relevant text in the document, or group the files in folders based on meaning.

## Royal Game of Ur Self-Reinforcement Learning AI

- I applied my knowledge in Reinforcement Learning by creating a bot that learns to play well at the game through self-play only.

## Wikipedia Recommendation System

- With my team, we created an online platform that recommends Wikipedia pages based on your preferences. I later improved the system by leveraging a lightweight sentence embedding model.

## Quantum Ground State Simulation

- I applied my knowledge in physics to write a program to find the ground state of a quantum system of interacting particles.

## Peak Detection in Noisy Signal

- I helped a friend in automating the detection and localization of peaks in noisy spectroscopic data on a user-friendly interface.

## tisext – Intelligent Time Series Extension

- A Python library that **extends periodic time series** by learning underlying patterns, even in noisy or interrupted signals. I tested it on a real-world dataset to **forecast product demand**.

## Telegram Bot

- I created a general-purpose bot that manages and interacts with Telegram users in real time.

## Mini-Hold'Em

- I understood and implemented **Counterfactual Regret Minimization (CFR)** to create a strong MiniPoker bot, despite hidden information playing a significant factor.

## Mancala Self-Reinforcement Learning AI

- A self-reinforcement learning AI for the game of Mancala. The agent uses 2-ply exhaustive search and a **neural network-based value function** to choose the best move. The value function leverages self-play, and is trained from scratch.